Pakistani Politician Image Classification Using Transfer Learning and Deep Convolutional Neural Networks

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***Abstract*—This project presents an AI-based image classification system for identifying Pakistani politicians using Deep Learning and Transfer Learning techniques. The dataset was collected using multiple online sources. Initially, images were downloaded manually using online libraries and Bing search engine; however, many irrelevant images were obtained. To improve dataset quality, the ImageEye browser extension was later used for efficient image collection and downloading.**

Two transfer learning models, ResNet50 and EfficientNetB0, were implemented and evaluated for politician classification. Data preprocessing, augmentation, and fine-tuning techniques were applied to improve model performance. Experimental results showed that the ResNet50 model achieved the best performance with approximately 89% test accuracy.

A Streamlit-based frontend application was also developed to allow users to upload images and receive real-time predictions.

***Index Terms*—Deep Learning, Transfer Learning, ResNet50, EfficientNetB0, Image Classification, CNN, Streamlit, TensorFlow**

I. Introduction

Artificial Intelligence and Deep Learning have significantly improved computer vision applications in recent years. Image classification systems are widely used in facial recognition, surveillance, healthcare, and biometric systems. Convolutional Neural Networks (CNNs) are among the most powerful Deep Learning architectures for image classification tasks.

This project focuses on developing an AI-based system capable of identifying Pakistani politicians from images. The proposed system uses transfer learning techniques with pre-trained CNN models including ResNet50 and EfficientNetB0.

The main objective of this project is to classify politician images into 16 different classes using Deep Learning techniques. A Streamlit-based frontend application was also developed to allow users to upload images and receive real-time predictions.

II. Dataset Collection and Preprocessing

The dataset used in this project consists of Pakistani politician images collected from online sources. Initially, images were downloaded manually using image libraries and Bing search engine. However, many irrelevant and duplicate images were collected during the early stages.

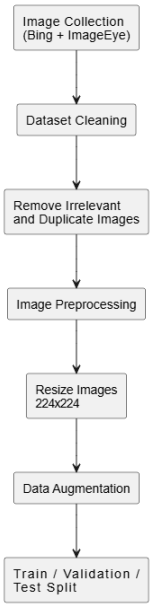

Fig. 1. Dataset Collection and Preprocessing Pipeline

To improve dataset quality and collection efficiency, the ImageEye browser extension was later used to download relevant politician images more effectively. After collection, irrelevant and corrupted images were manually removed.

The final dataset contains 16 politician classes with approximately 100 to 135 images per class.

The dataset was divided into:

- 75% Training Data
- 15% Validation Data
- 10% Testing Data

Image preprocessing and augmentation techniques were

applied including:

- Image resizing to 224x224
- Rotation
- Horizontal flipping
- Zoom augmentation
- Brightness adjustment
- Width and height shifting

III. Methodology

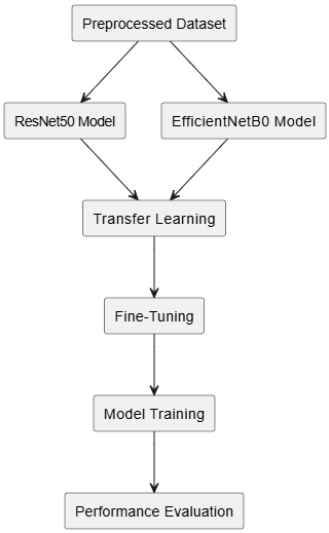

Fig. 2. Model Training Pipeline

The proposed system uses transfer learning-based Deep Learning models for image classification.

A. ResNet50

ResNet50 is a Deep Residual Network containing 50 layers. It uses residual learning and skip connections to reduce vanishing gradient problems in deep networks.

In this project, the ImageNet pre-trained ResNet50 model was used with transfer learning. The top classification layer was removed and replaced with:

- Global Average Pooling Layer
- Dropout Layer
- Dense Softmax Classification Layer

Initially, the base model layers were frozen to preserve pre-trained ImageNet features. Later, fine-tuning was performed by unfreezing the last layers of the network.

B. EfficientNetB0

EfficientNetB0 is a lightweight CNN architecture that balances network depth, width, and resolution efficiently.

EfficientNetB0 was implemented using transfer learning and evaluated on the same dataset.

C. Frontend Development

A Streamlit-based web application was developed for real-time prediction. Users can upload politician images through the frontend interface, and the trained model predicts the politician name with confidence score.

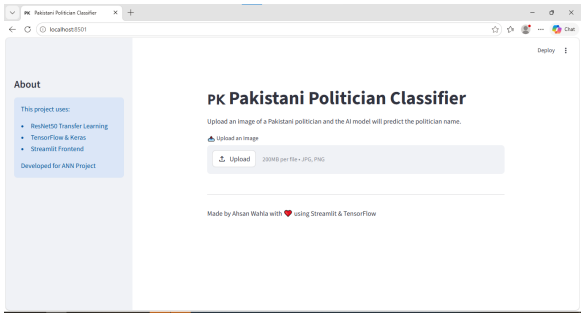

Fig. 3. Streamlit Frontend Interface

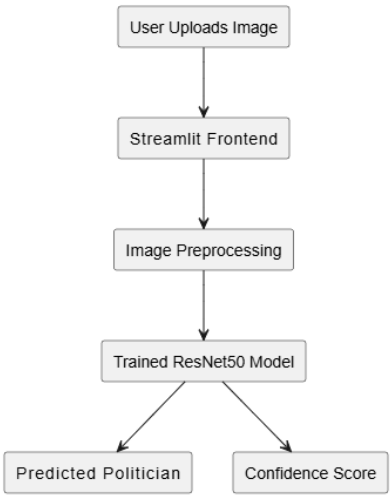

Fig. 4. Inference System Architecture

IV. Training Strategy

The training process was performed using transfer learning techniques. Initially, the pre-trained convolutional base layers were frozen to preserve ImageNet feature representations. Only the custom classification layers were trained during the initial training phase.

Data augmentation techniques including rotation, horizontal flipping, zooming, brightness adjustment, width shifting, and height shifting were applied to improve model generalization and reduce overfitting.

After the initial training stage, fine-tuning was performed by unfreezing the last layers of the ResNet50 model.

The Adam optimizer and categorical cross-entropy loss function were used during training.

V. Experimental Results

Both ResNet50 and EfficientNetB0 models were evaluated on the testing dataset.

A. ResNet50 Results

- Training Accuracy: 95%
- Validation Accuracy: 89%
- Test Accuracy: 89.16%
- Test Loss: 0.328

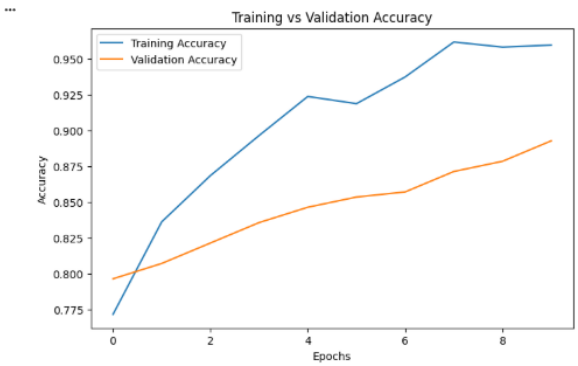

Fig. 5. Training and Validation Accuracy Graph

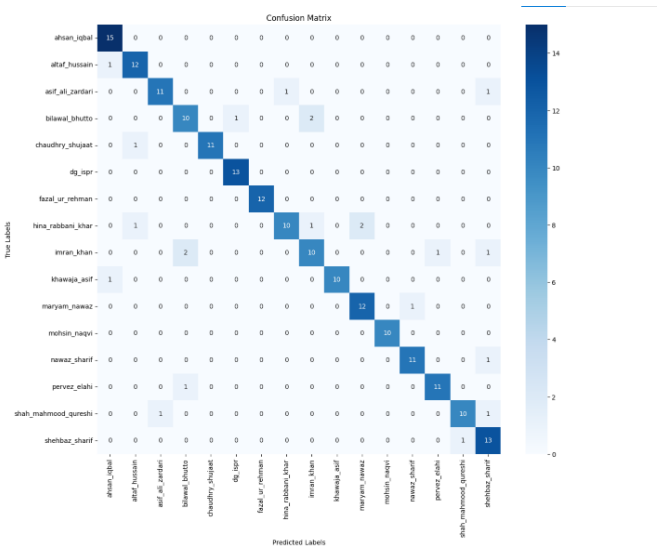

Fig. 6. Confusion Matrix of ResNet50 Model

B. *EfficientNetB0 Results*

- Test Accuracy: 77.83%
- Test Loss: 0.944

VI. Results Comparison

TABLE I
Model Comparison

Model	Test Accuracy	Test Loss
ResNet50	89.16%	0.328
EfficientNetB0	77.83%	0.944

Experimental results showed that ResNet50 significantly outperformed EfficientNetB0 in terms of accuracy and classification reliability.

VII. Misclassification Analysis

Some misclassifications were observed during testing because of low-resolution images, side-face angles, multiple people in images, and visually similar politician features.

VIII. Challenges Faced

- Collecting high-quality politician images was difficult because many irrelevant images were downloaded initially from Bing search results.
- Dataset imbalance and duplicate images affected model training during early stages.
- Some images contained multiple people, unclear faces, or poor lighting conditions.
- Training deep learning models required high computational resources and longer execution time.
- Fine-tuning CNN models required careful experimentation to avoid overfitting.

IX. Findings and Discussion

The experimental results demonstrated that transfer learning significantly improved classification performance even with a relatively limited dataset. ResNet50 achieved the best overall performance with approximately 89% testing accuracy and generalized effectively on unseen images.

EfficientNetB0 achieved lower accuracy but required fewer trainable parameters and computational resources.

X. Conclusion

This project presented a Deep Learning-based Pakistani politician image classification system using transfer learning techniques. Two CNN architectures, ResNet50 and EfficientNetB0, were implemented and evaluated.

Experimental results showed that ResNet50 achieved the best performance with approximately 89% test accuracy.

Future improvements may include increasing dataset size, implementing face detection techniques, and deploying the system on cloud platforms for large-scale real-time usage.

Acknowledgment

The author would like to thank Sir Asif Ameer for his valuable guidance and supervision throughout this Artificial Neural Networks project.

References

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